STRUCTURAL ANALYSIS AND STEEL DESIGN OF A THREE-STORY STEEL BUILDING

Software: SAP2000 v20.0.0
Design Standard: AISC 360-16 (LRFD)

PROJECT PORTFOLIO REPORT

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Date: 10 Oct, 2026

1. Project Overview

This project presents the structural analysis and steel design of a hypothetical three-story steel building using SAP2000 v20.0.0. The project was developed as an educational structural engineering portfolio to demonstrate structural modeling, load application, analysis and steel member design using a commercial structural analysis and design program.

The building is modeled as a steel-framed structure with a plan dimension of 80 ft × 40 ft and a total height of 36 ft, consisting of three stories with a story height of 12 ft. ASTM A992 structural steel is specified for the steel members.

The steel frame design is evaluated using the Load and Resistance Factor Design (LRFD) method in accordance with AISC 360-16. The project includes the review of structural response, support reactions, joint displacements and member demand-to-capacity ratios obtained from SAP2000.

The purpose of this project is to demonstrate an organized structural engineering workflow and document the analysis and design results for academic and professional portfolio use. The model is hypothetical and is not intended for construction without further verification and professional review.

2. Building Description

The structure considered in this project is a hypothetical three-story steel building with a rectangular plan. The structural system consists of steel beams and columns arranged to provide a load-carrying framework for gravity and lateral loading.

The building has overall plan dimensions of 80 ft in the longitudinal direction and 40 ft in the transverse direction. Each story has a height of 12 ft, giving a total building height of 36 ft.

The structural members are modeled in SAP2000 using steel frame elements. The member sections are assigned according to the selected design trial and the model is analyzed to evaluate its structural response under the defined loading conditions.

The principal building parameters are summarized below.

Parameter	Description
Building type	Office building
Structural system	Steel-framed structure
Number of stories	3
Story height	12 ft
Total height	36 ft
Plan dimensions	80 ft × 40 ft
Structural material	ASTM A992 steel
Analysis software	SAP2000 v20.0.0

3. Materials and Design Standards

3.1 Structural Material

ASTM A992 structural steel is specified for the primary steel framing members in the analytical model. The material properties assigned in the model are summarized below.

Property	Value
Specified yield strength, (Fy)	50 ksi
Specified tensile strength, (Fu)	65 ksi
Modulus of elasticity, (E)	29,000 ksi

3.2 Design Standard

The steel members are evaluated using the American Institute of Steel Construction (AISC) Specification for Structural Steel Buildings, AISC 360-16, using the Load and Resistance Factor Design (LRFD) method.

LRFD evaluates structural members by comparing the required strength under factored loads with the available design strength determined in accordance with the applicable specification provisions.

3.3 Analysis Software and Units

SAP2000 version 20.0.0 is used for structural modeling, analysis and steel frame design. The model uses kip, ft and °F as its reported unit system, with steel stresses and material properties expressed in ksi where applicable.

4. Structural Modeling

4.1 Model Development

The three-story steel building was modeled in SAP2000 v20.0.0 using a three-dimensional analytical model. Steel beams and columns were represented by frame elements connected at structural joints. The model was developed to represent the building's geometry and provide a basis for structural analysis and steel member design.

4.2 Structural Members

The trial steel sections used in the model include W12×79 columns, W18×35 primary beams, and W16×40 secondary beams. These sections were evaluated through the steel frame design process and the resulting demand-to-capacity ratios were reviewed to identify the governing member.

4.3 Supports and Connectivity

Structural supports and member connectivity were defined in the SAP2000 model to represent the intended load-transfer system. The analysis results depend on the assigned restraints, joint connectivity, member properties and loading conditions. These model inputs should be checked against the intended structural configuration before the results are accepted.

4.4 Analysis Procedure

After the model geometry, material properties, member sections and loading definitions were assigned, the structural analysis was run in SAP2000. The resulting analysis outputs were used to review support reactions, joint displacements and the response of the steel frame. Steel member design results were then examined using the AISC 360-16 design settings.

5. Loads and Load Combinations

5.1 Applied Loads

The structural model considers gravity loads and lateral wind loads to evaluate the response of the steel-framed building. The load patterns defined for the project include the following:

Load pattern	Description
DEAD (D)	Dead load from structural self-weight and other assigned permanent loads
LIVE (L)	Occupancy-related imposed loads
WIND-X (WX)	Wind loading acting along the global X direction
WIND-Y (WY)	Wind loading acting along the global Y direction

The magnitude and distribution of these loads depend on the model inputs and the assumptions adopted for the hypothetical building. The assigned values should be verified against the SAP2000 model before final submission.

5.2 Load Combinations

Load combinations are used to evaluate the structure under different combinations of loading effects. The steel design results include the combination:

1.2D + 1.0L + 1.0WY(-)

This combination produced the highest reported PMM ratio among the exported steel design results. The PMM ratio identifies the combined effect of axial force and biaxial bending for the member under the selected design combination.

5.3 Purpose of Load Combinations

Different load combinations can produce different governing effects in columns and beams. Therefore, the governing member is identified from the design results rather than assuming that one combination controls every member or every type of structural response.

6. Structural Analysis Results

6.1 Analysis Overview

The structural model was analyzed using SAP2000 to evaluate its response to the defined loading conditions. The analysis outputs include the deformed shape of the structure, support reactions, and joint displacements. These results provide information about the overall structural response and the forces transferred to the supports.

6.2 Deformed Shape

The deformed shape was reviewed to observe the structural response under lateral wind loading. The WIND-X deformed-shape view was captured for project documentation. The deformation pattern should be checked for consistency with the applied loading direction, support conditions, and expected structural behavior.

6.3 Support Reactions

Support reactions were exported from SAP2000 and saved in the file *Support_Reactions.xlsx*. These results represent the forces and moments developed at the model's restrained joints. They can be used to review the transfer of applied loads to the building supports.

6.4 Joint Displacements

Joint displacement results were exported and saved in *Joint_Displacements.xlsx*. These results contain translational and, where applicable, rotational response values at the model joints. Displacements under wind loading should be reviewed against the applicable serviceability criteria before conclusions are made about lateral drift or overall performance.

7. Steel Design Results

7.1 Design Evaluation

Steel frame design was performed in SAP2000 using AISC 360-16 with the LRFD method. The exported steel design results include the PMM ratio, which evaluates the interaction of axial force and biaxial bending for the applicable member design checks.

7.2 Governing Member

The exported design results identified a W12×79 column as the member with the highest reported PMM ratio. The governing result is summarized below.

Parameter	Result
Member section	W12×79
Member type	Column
PMM ratio	0.901156
Governing load combination	1.2D + 1.0L + 1.0WY(-)

7.3 Interpretation of Results

The governing PMM ratio of 0.901156 is below the interaction limit of 1.0. This indicates that the member satisfies the reported PMM interaction check for the governing combination in the exported results.

However, this result alone does not establish that all members and all applicable requirements are satisfied. The complete design assessment should also consider relevant stability, slenderness, serviceability, connection and other applicable code requirements.

7.4 Design Results Documentation

The exported design table is saved in *Steel_Design_PMM_Ratios.xlsx*. This workbook provides a record of the reported member design ratios and supports the review of the governing design result.

8. Conclusion and Limitations

8.1 Conclusion

This project demonstrated the structural modeling, analysis and steel member design workflow for a hypothetical three-story steel office building using SAP2000 v20.0.0. The steel frame design was evaluated using AISC 360-16 and the LRFD method.

The exported steel design results identified a W12×79 column as the governing member, with a maximum reported PMM ratio of 0.901156 under the load combination 1.2D + 1.0L + 1.0WY(-). This ratio is below 1.0 and satisfies the reported PMM interaction check for that member and combination.

Support reactions, joint displacements and the WIND-X deformed shape were also documented as part of the project workflow. These outputs provide a basis for further review of the building's structural response and design assumptions.

8.2 Limitations

This project is an educational portfolio exercise based on a hypothetical building. The reported results depend on the model geometry, material properties, support conditions, applied loads, load combinations and software design settings.

The reported PMM ratio does not independently verify every member, connection, stability requirement, drift limit, foundation condition or other applicable design criterion. Further checks and independent engineering review are required before the model could be considered for any real-world construction application.

The model and report should therefore be treated as documentation of an educational structural analysis and design exercise, not as a construction-ready or certified engineering design.